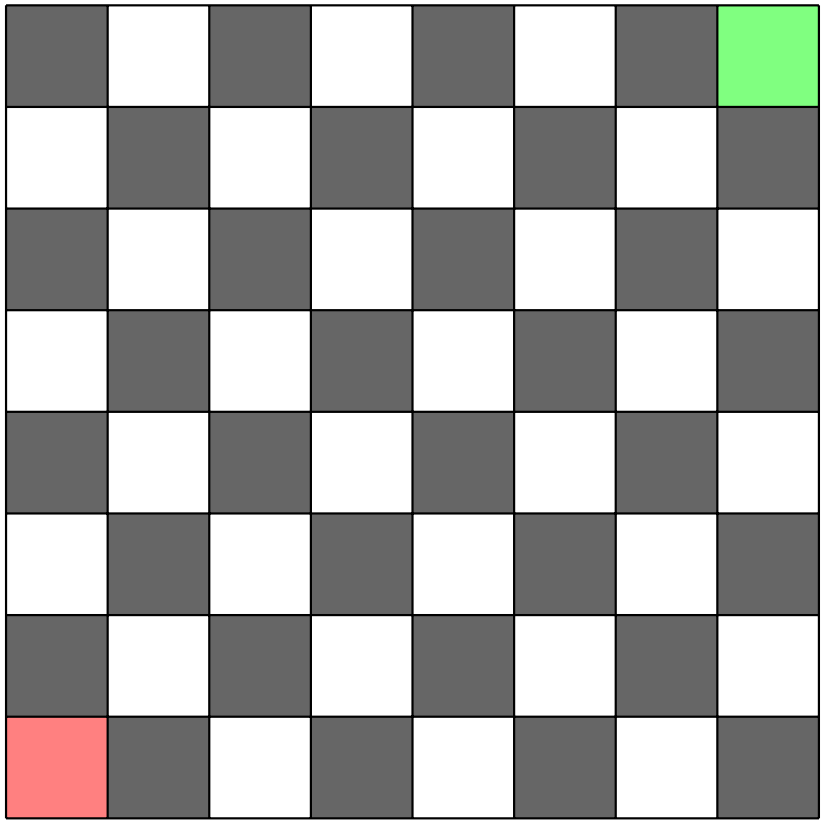
CSE 101M – Fall 2025 – Lecture 5: Combinatorics, Counting, and Induction

Mathematical Thinking for Computer Science

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TAs: Steven,Yiyuan

October 13, 2025

How many "monotone" rook moves?



Q: A rook at a corner wants to reach the opposite corner (no backtracking, so non-decreasing coordinates). Count how many ways there are.

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- Next, we give examples via:
  1. combinatorial arguments,
  2. algebra (binomial/multinomial theorems),
  3. induction (weak or strong).

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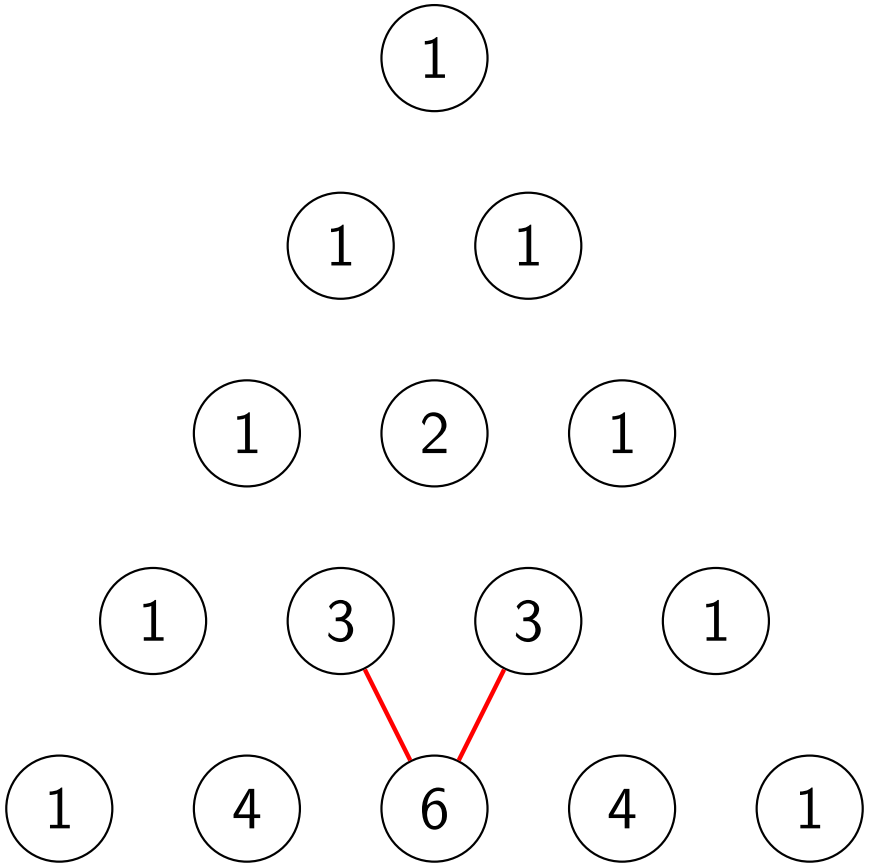
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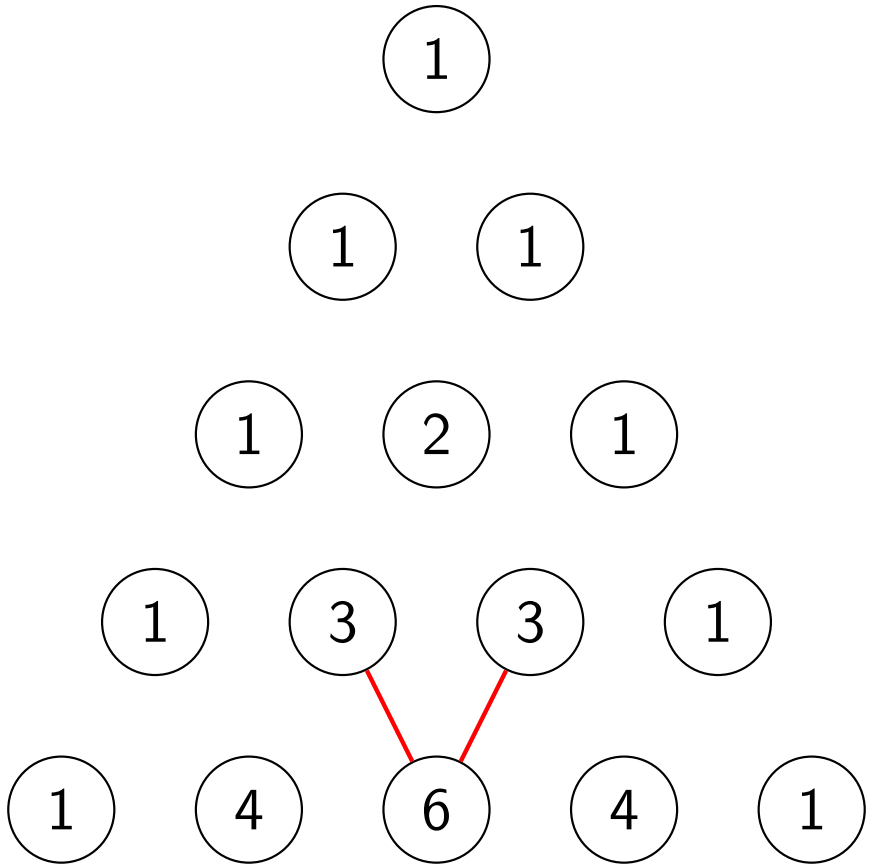
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Therefore $\displaystyle\sum_{k=1}^{n} \binom{n}{k} = 2^n - 1.$

Pascal's Triangle



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Each number is the sum of the two numbers directly above it. Used for:

$$(x+y)^2 = x^2 + 2xy + y^2, \text{ and for } \binom{4}{2} = \binom{3}{1} + \binom{3}{2} = 6.$$

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Add the two cases: $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$.

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Hence $\sum_{k=0}^{n} k\binom{n}{k} = n\,2^{n-1}$.

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Therefore $S(n+1) = (n+1)2^n$.

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$$= \sum_{k=0}^{n+1} k \left[\binom{n}{k} + \binom{n}{k-1} \right] \qquad \text{(Pascal's formula)}$$

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Handle $(**)$ *by re-indexing.*

$$(**) = \sum_{j=-1}^{n} (j+1)\binom{n}{j} = \sum_{j=0}^{n} j\binom{n}{j} + \sum_{j=0}^{n} \binom{n}{j} = S(n) + 2^n$$

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Algebraic proof. Expand $(1-1)^n = \sum_{k=0}^{n}\binom{n}{k}1^{n-k}(-1)^k = 0$.

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Therefore the coefficient of $x^{n-k}y^k$ is $\binom{n}{k}$, proving the identity. □

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- LHS: first pick r from the m-block and $k-r$ from the n-block, then sum over all r.

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- Using Pascal's rule or identities outside their valid ranges (e.g., negative indices without care).

Practice more in HW3

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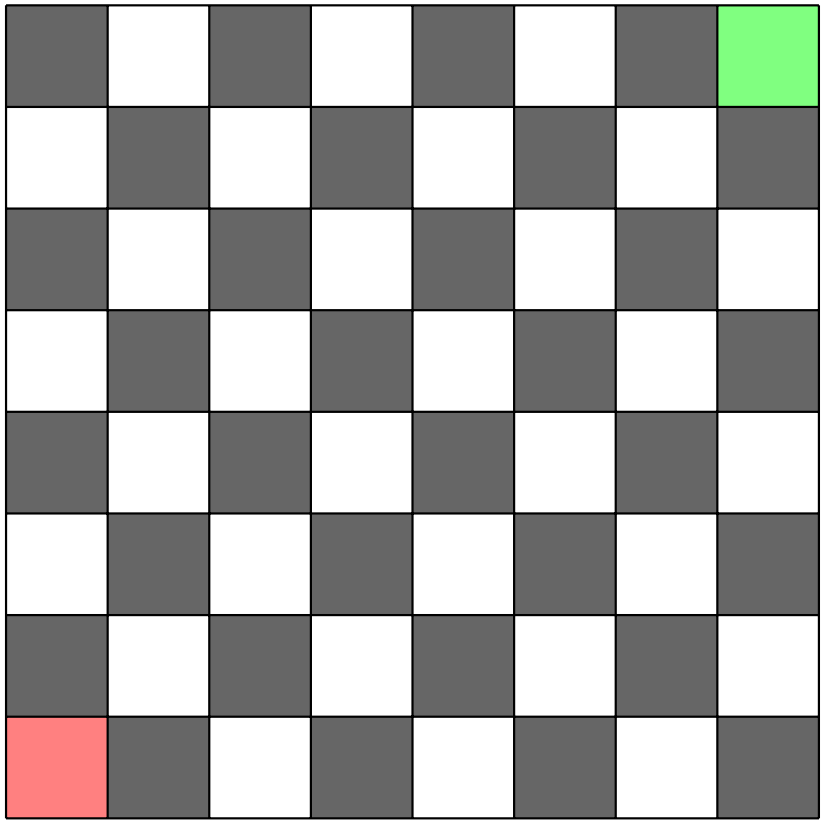
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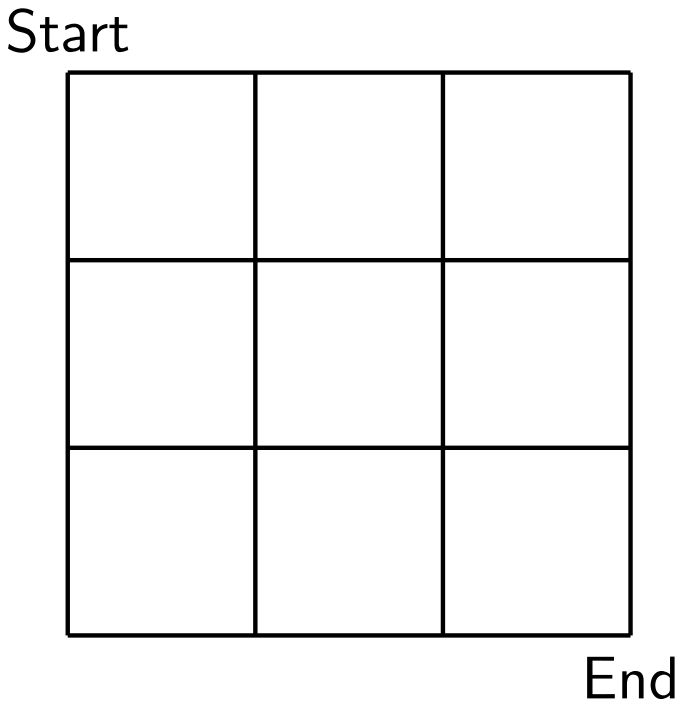
1. $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$.
2. For $n \geq 1$: $\sum_{k=1}^{n} k^2 \binom{n}{k} = n(n+1)2^{n-2}$.

How many "monotone" rook moves?

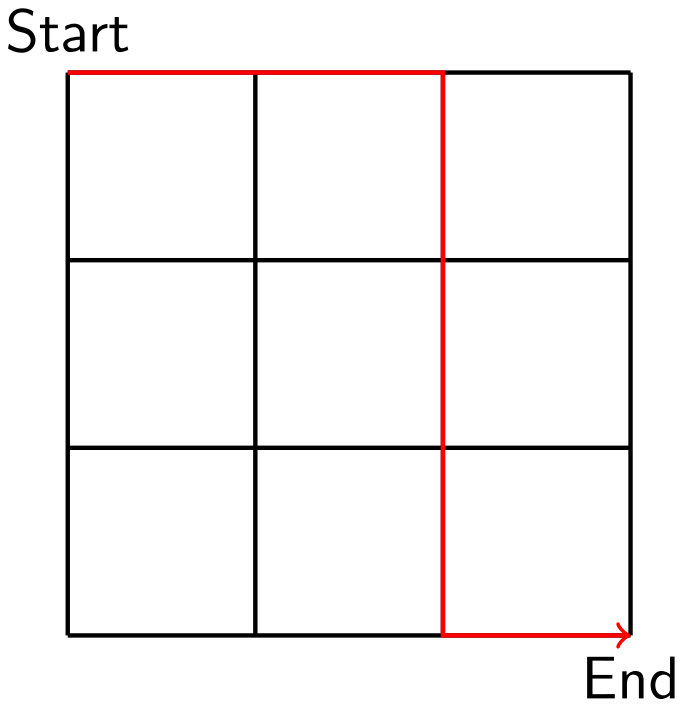


Q: A rook at a corner wants to reach the opposite corner (no backtracking, so non-decreasing coordinates). Count how many ways there are.

Paths in a 3×3 Grid



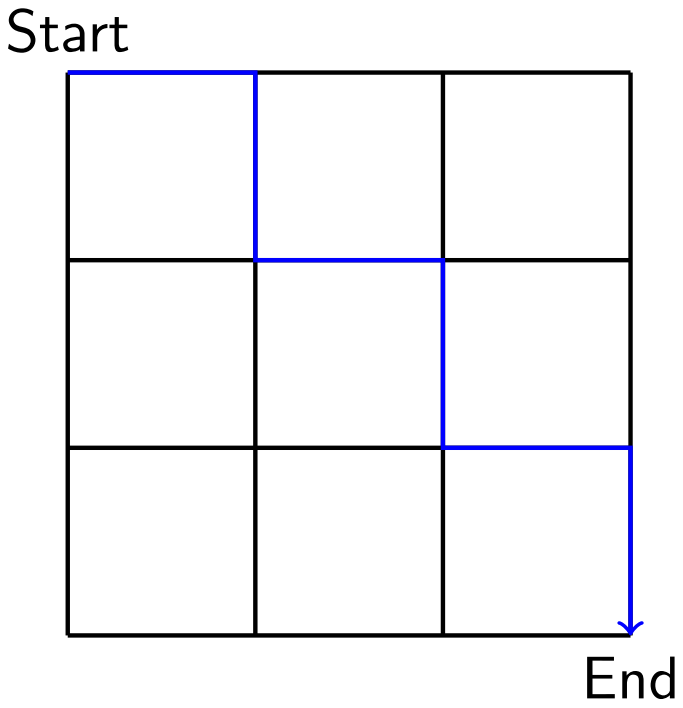
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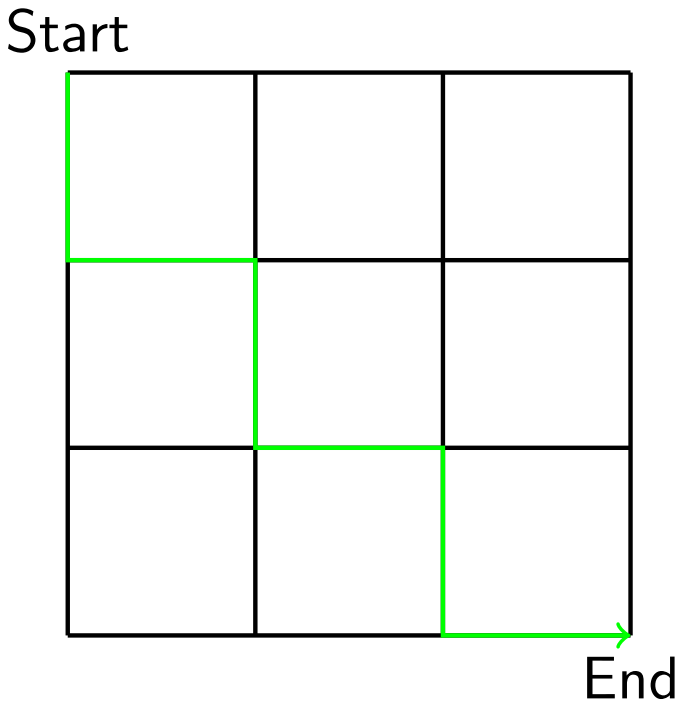
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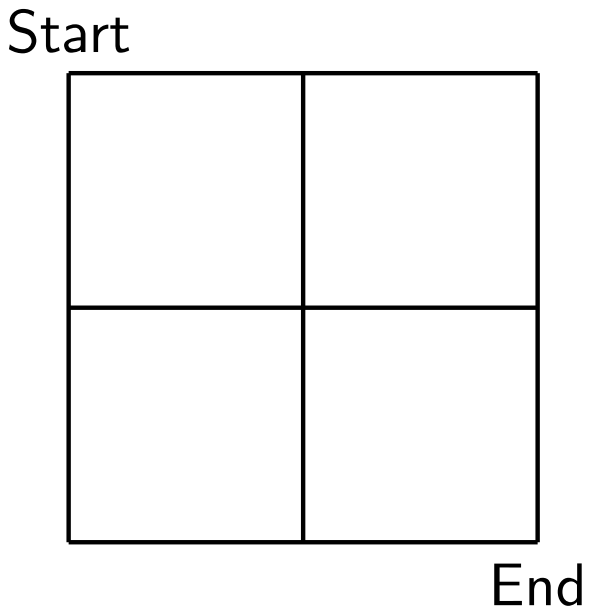
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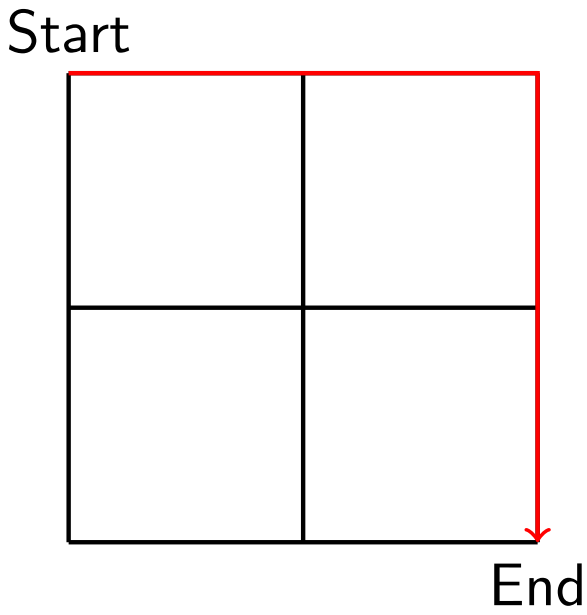
Observation: For a 3×3 grid, the total number of paths is

$$\binom{3+3}{3} = 20$$

Paths in a 2×2 Grid

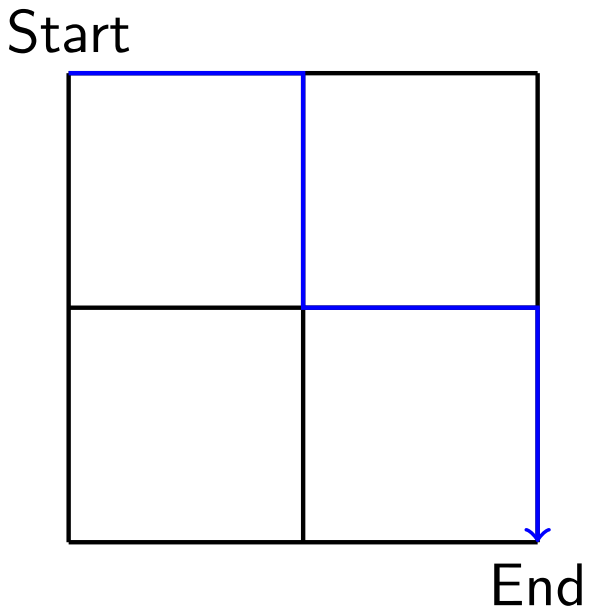


Paths in a 2×2 Grid



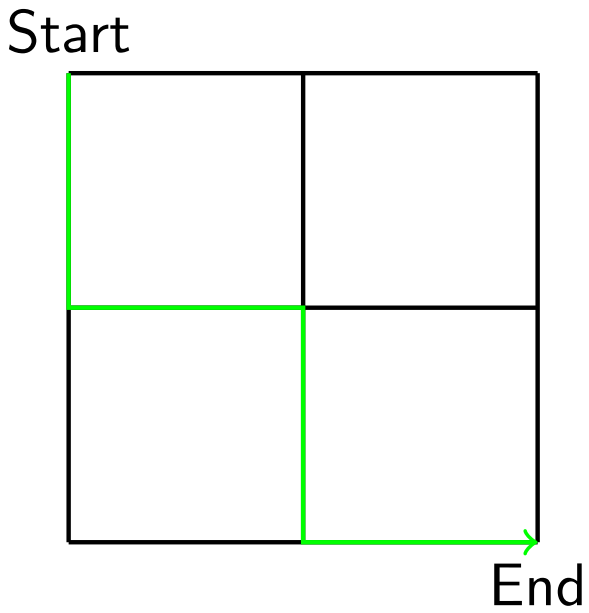
Observation: For a 2×2 grid, total paths $= \binom{2+2}{2} = 6$ (remaining 2 paths can be added similarly).

Paths in a 2×2 Grid



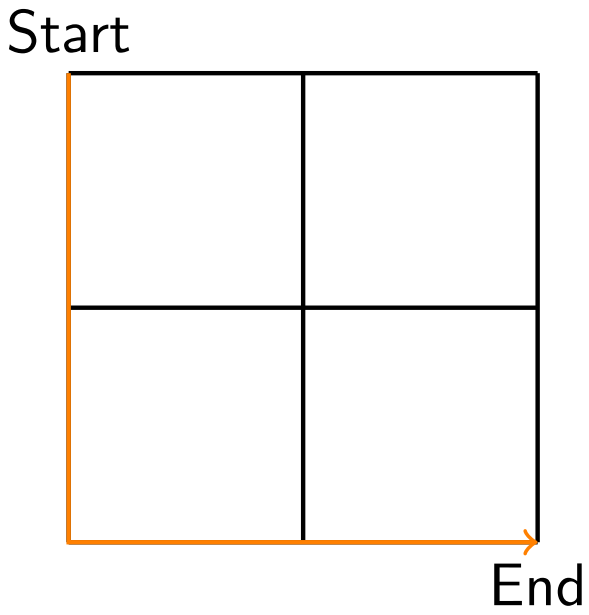
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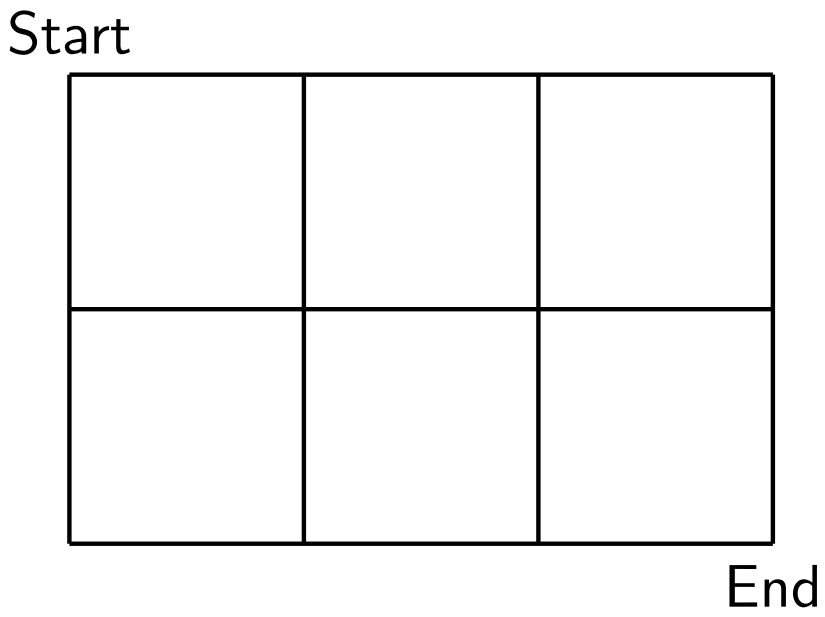
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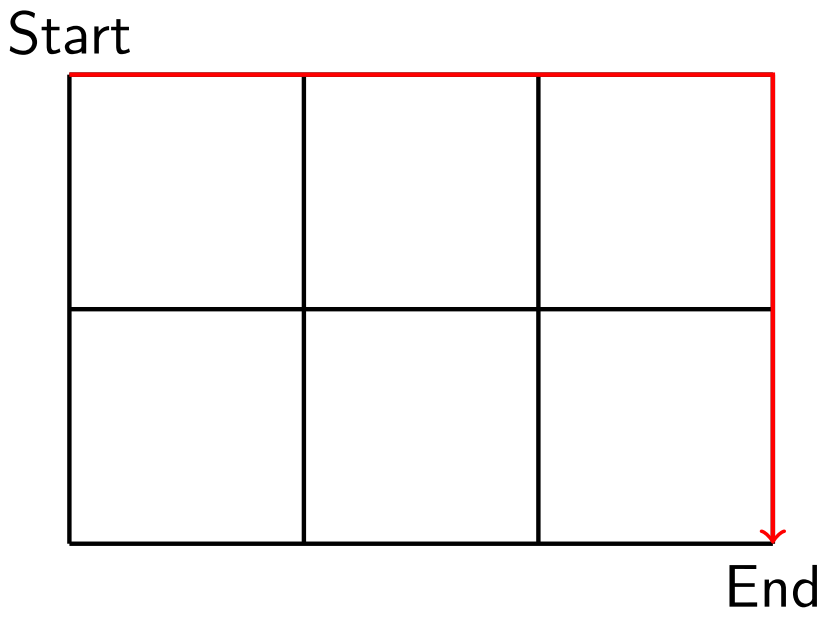


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Paths in a 2×3 Grid

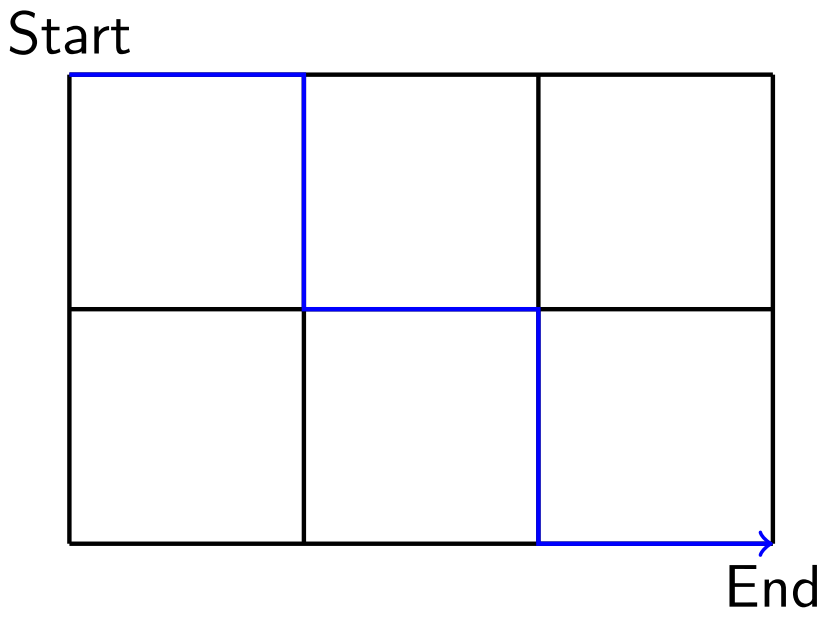


Paths in a 2×3 Grid



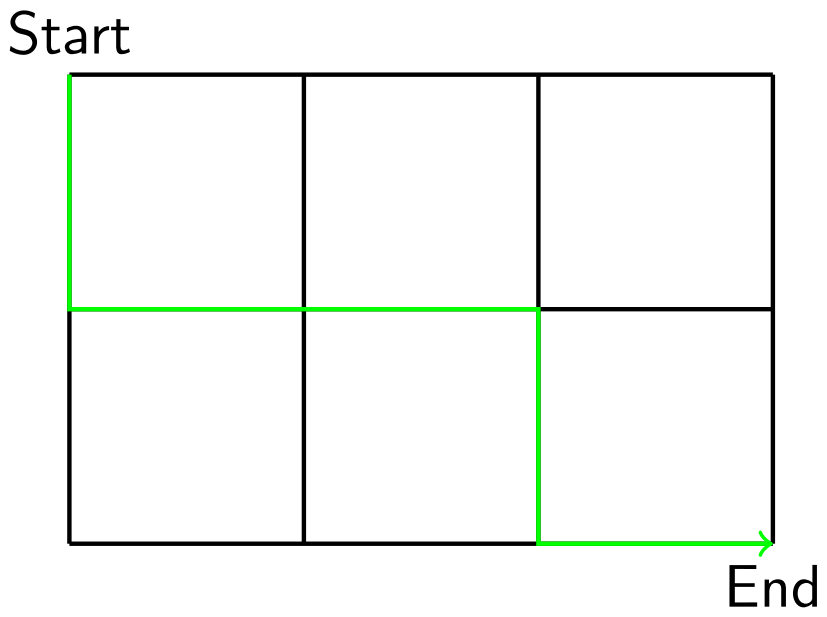
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Paths in an $m \times n$ Grid - Induction Proof

Statement: Number of paths from top-left to bottom-right in an $m \times n$ grid:

$$P(m, n) = \binom{m+n}{m} = \binom{m+n}{n}$$

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$$P(m, 0) = P(0, n) = 1$$

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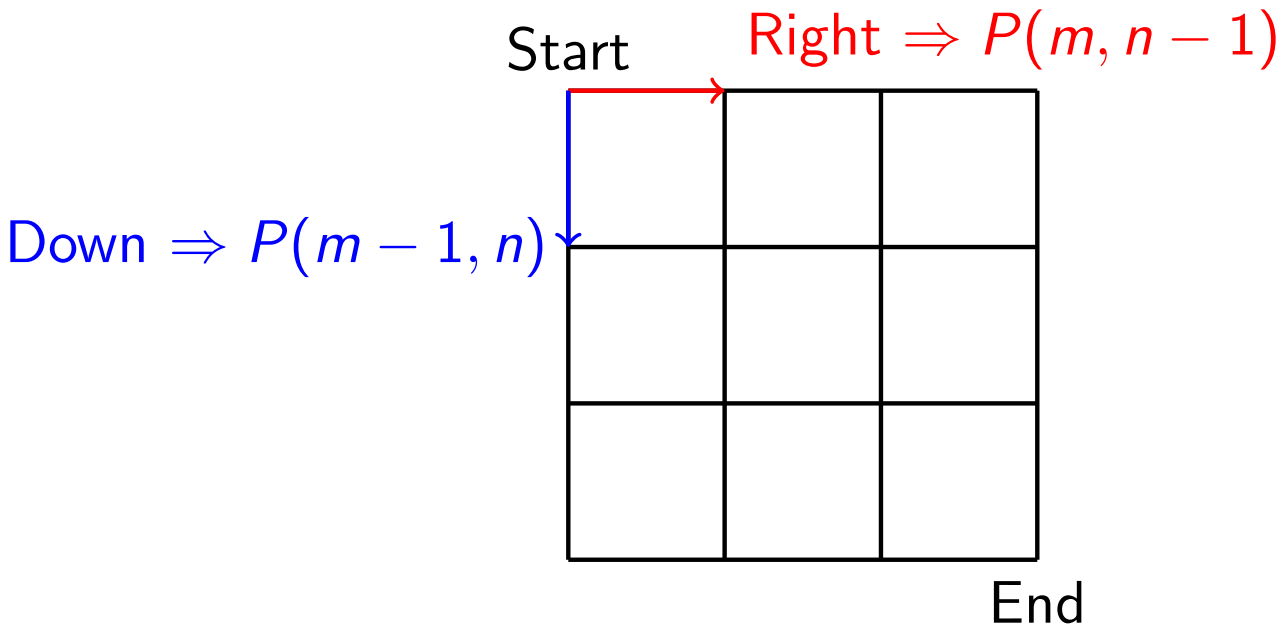
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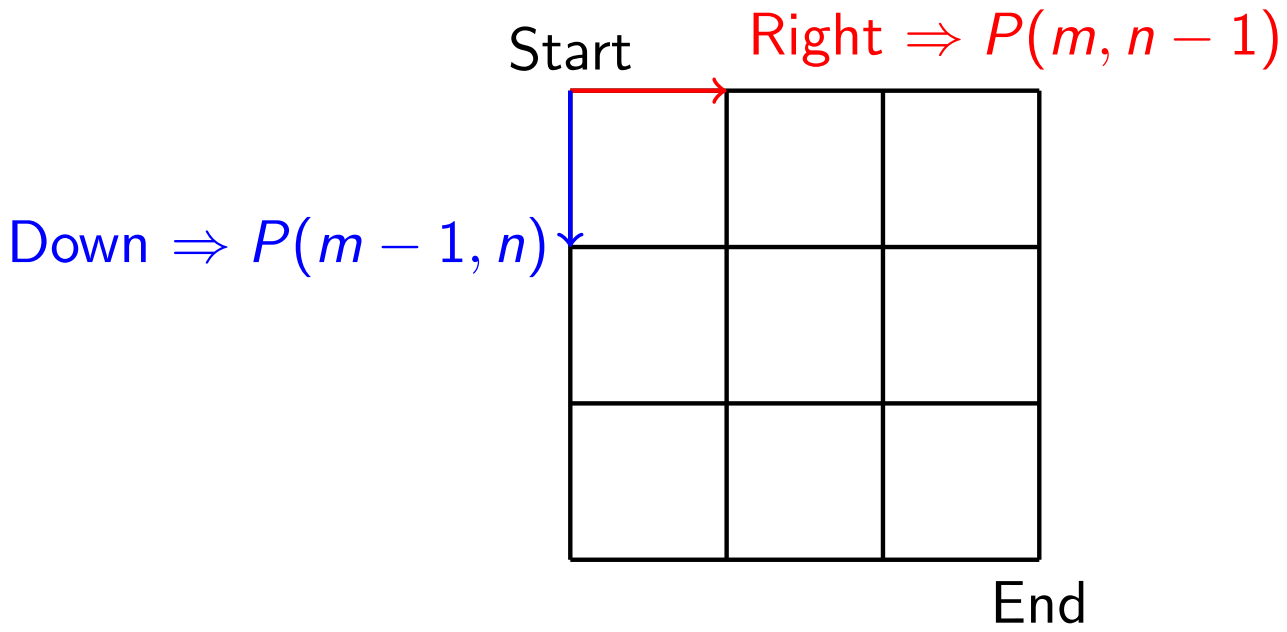
Induction Hypothesis: Assume true for all grids with $m + n = k$.

Induction Step: Now say we have $m + n = k + 1$, what can the first move be?

The Next Move



The Next Move

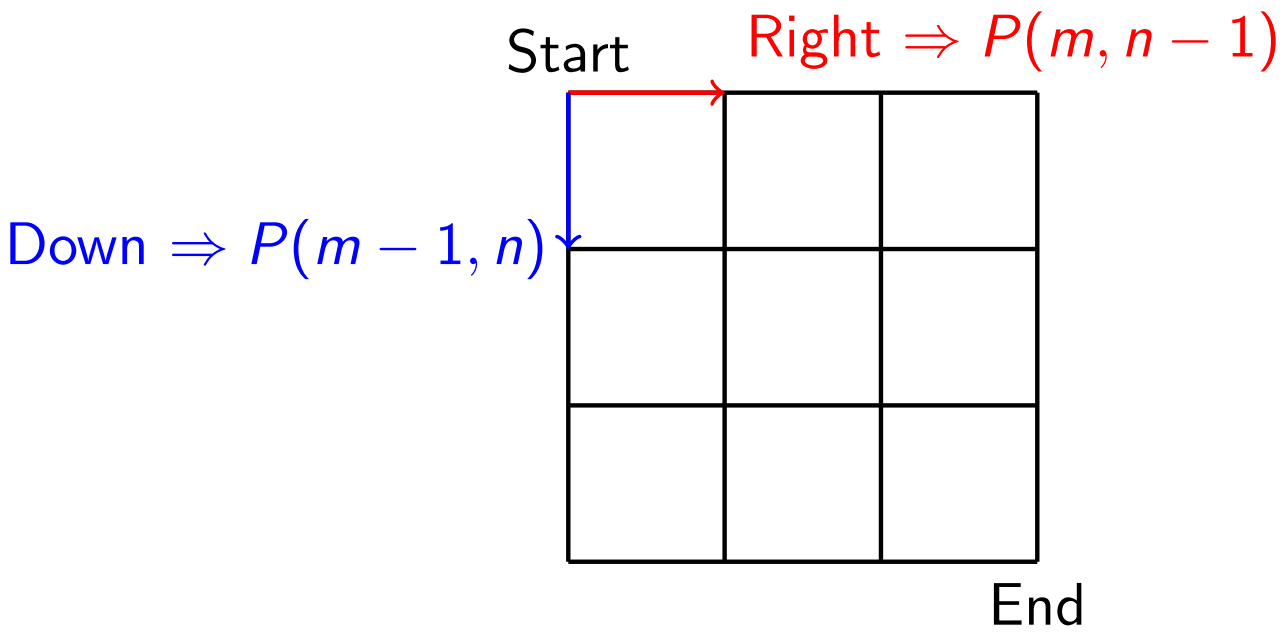


By induction hypothesis (using Pascal's formula):

$$P(m,n) = P(m-1,n) + P(m,n-1) \implies$$

$$P(m,n) = \binom{m+n-1}{m-1} + \binom{m+n-1}{m} = \binom{m+n}{m}$$

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Conclusion: Formula holds for all $m, n \geq 0$.

CSE 101M – Fall 2025 – Lecture 6
Basics of Logic and Reasoning
Mathematical Thinking for Computer Science

PI: Vaggos Chatziafratis
TAs: Steven,Yiyuan

October 15, 2025

What is Propositional Logic?

- A **proposition** is a statement that is either **true (T)** or **false (F)**, but not both.

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- A **proposition** is a statement that is either **true (T)** or **false (F)**, but not both.
- Examples:
  - "$2 + 3 = 5$" $\rightarrow$ True
  - "Paris is the capital of Italy." $\rightarrow$ False

Logical Connectives

Operation	**Symbol**	**Name**
Negation	$\neg p$	"not p"
Conjunction	$p \wedge q$	"p and q"
Disjunction	$p \vee q$	"p or q"
Conditional	$p \rightarrow q$	"if p then q"
Biconditional	$p \leftrightarrow q$	"p if and only if q"

Truth Tables for Basic Connectives

Negation:

p	$\neg p$
T	F
F	T

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T	T	T
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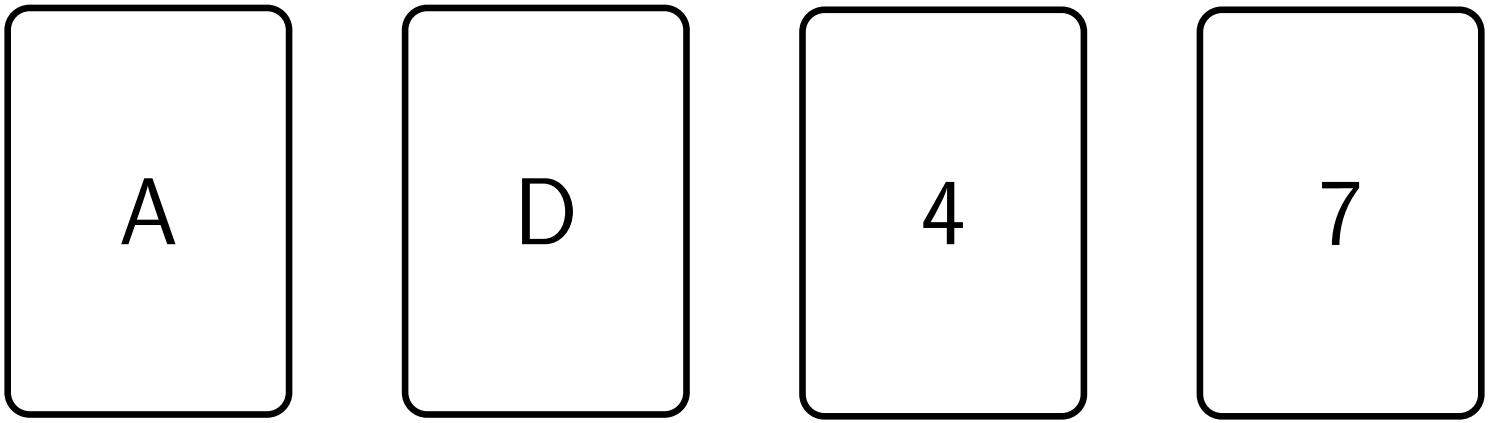
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Implication:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

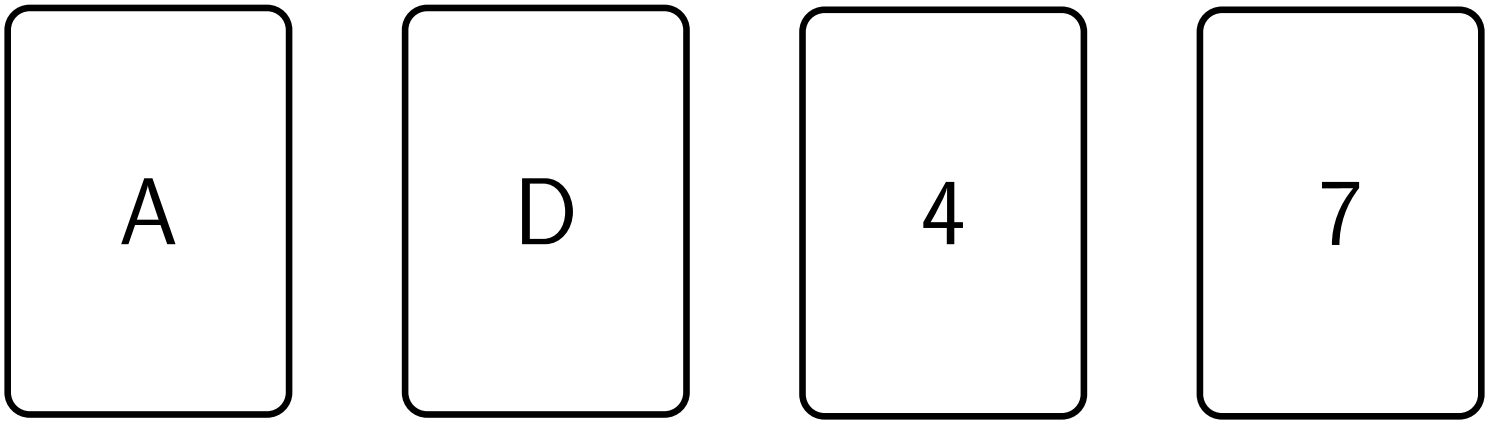
Wason Selection Task

Each card has a **letter** on one side and a **number** on the other.



Wason Selection Task

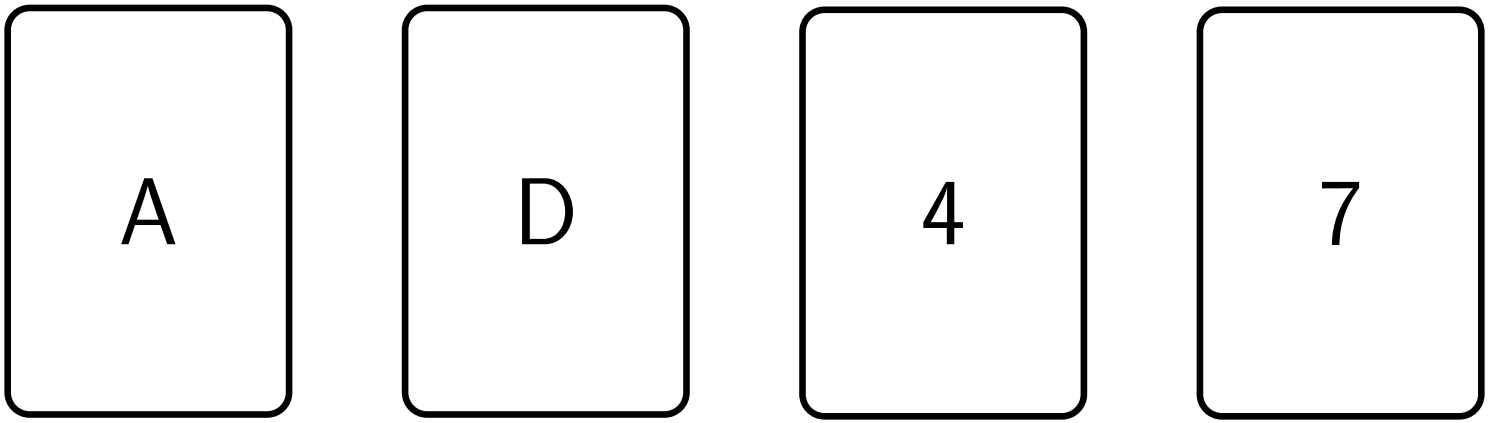
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Rule to test: "If a card has a vowel on one side, then it has an even number on the other side."

Wason Selection Task

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Rule to test: "If a card has a vowel on one side, then it has an even number on the other side."

Q: Which card(s) *must* you turn over to test the rule?

Formalizing the rule

Let $p =$ "vowel on this side", $q =$ "even number on the other side."

Target Proposition

$$p \to q \quad \text{(If vowel, then even.)}$$

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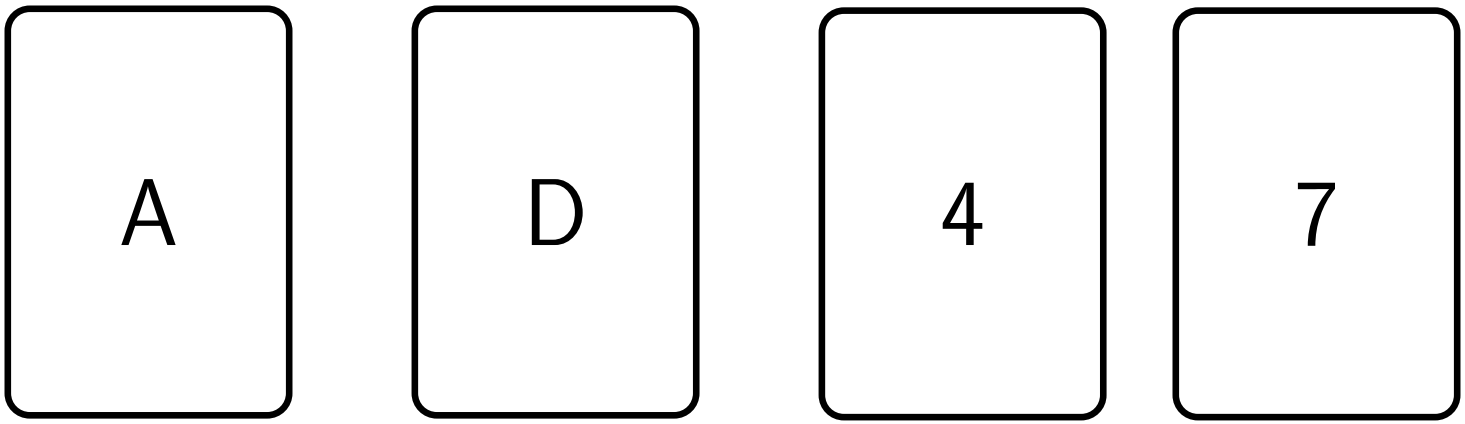
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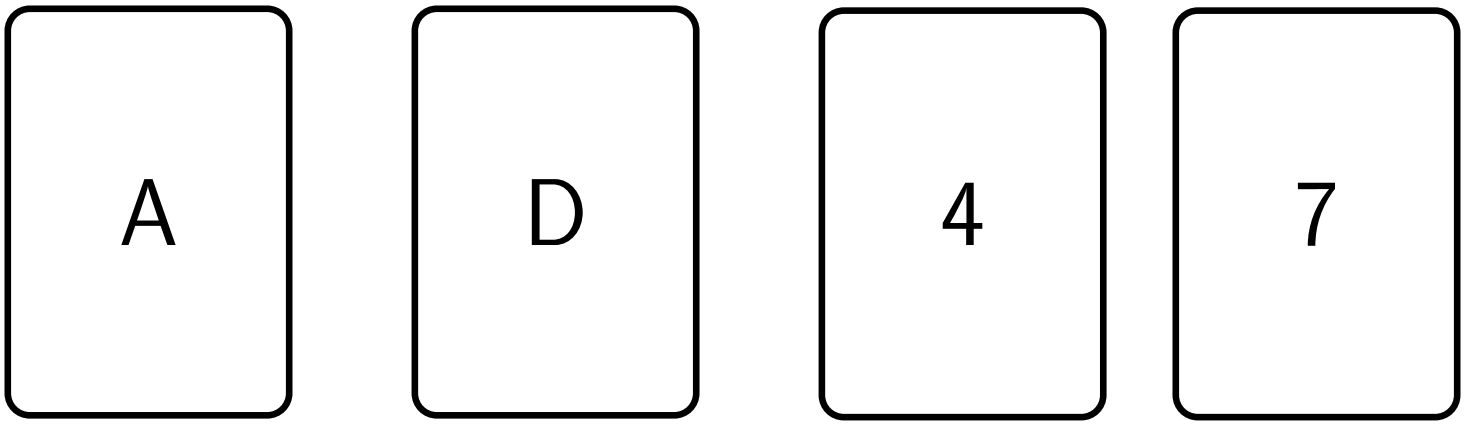
$$p \to q \quad \text{(If vowel, then even.)}$$

- To **falsify** $p \to q$, we need a case where p is true but q is false: $(p \wedge \neg q)$.
- So we must check:
  - all **p**-cards (vowel showing), and
  - all $\neg$**q**-cards (odd number showing).

Mapping the four cards

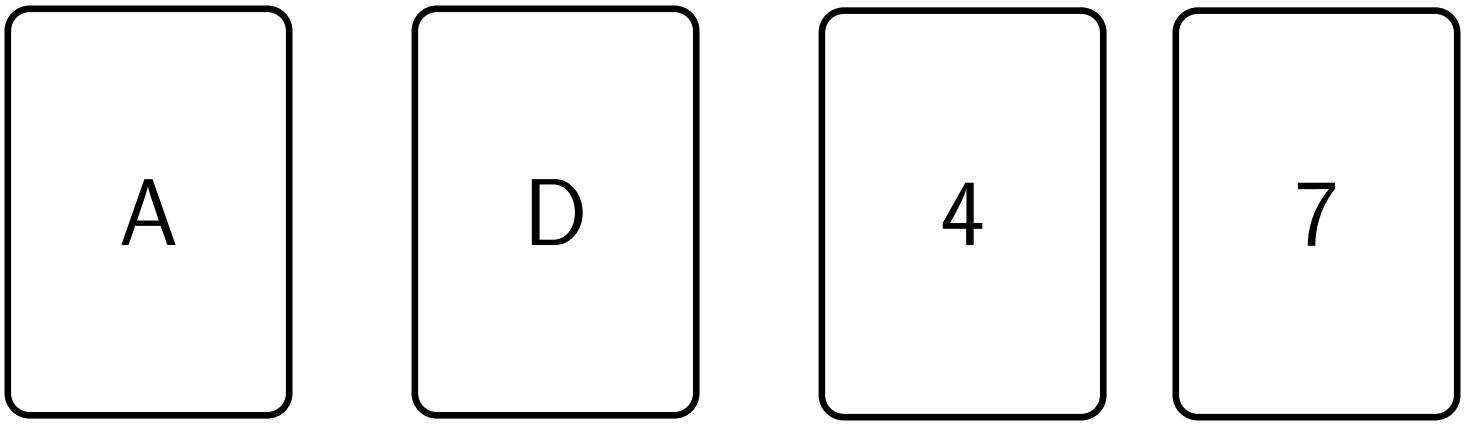


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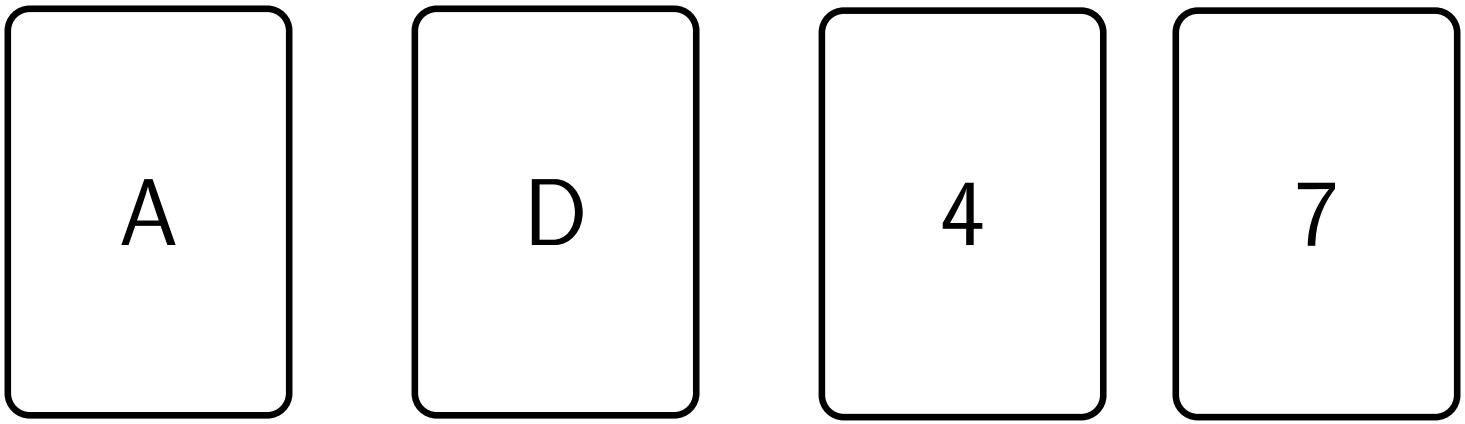
- Check **A** (p): it can falsify if the other side is odd ($\neg q$).

Mapping the four cards



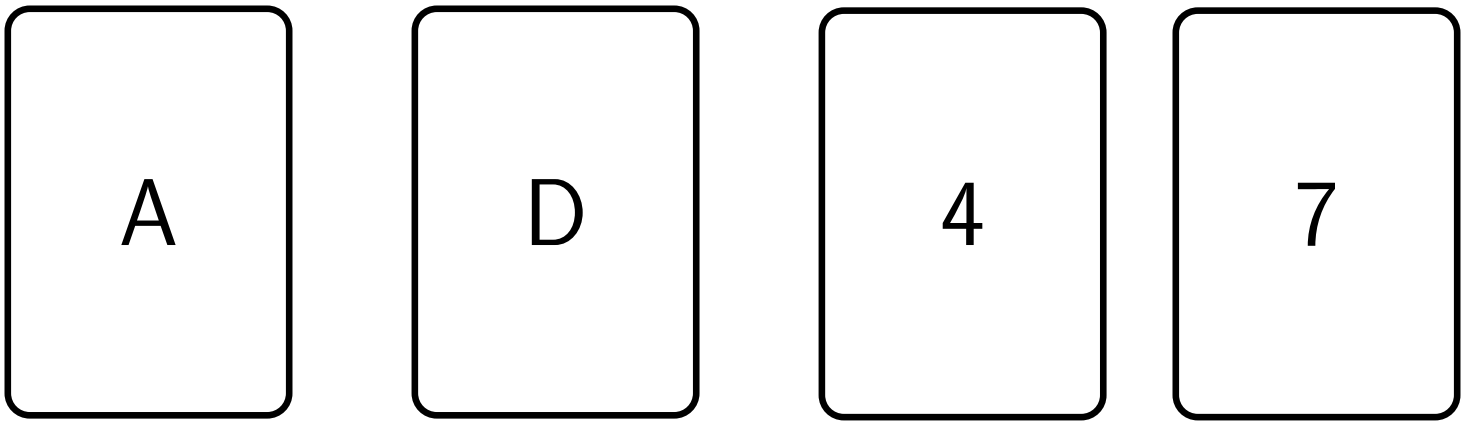
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Mapping the four cards



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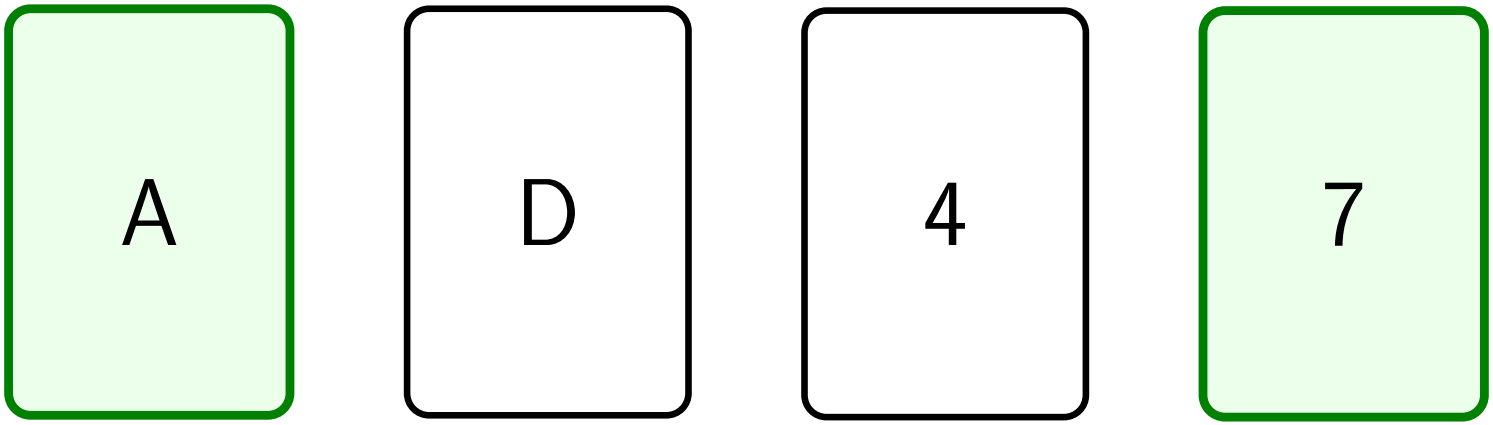
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- *Do not need* to check **D** ($\neg p$): whether even or odd, $\neg p$ cannot produce $p \wedge \neg q$.

A compact falsification table

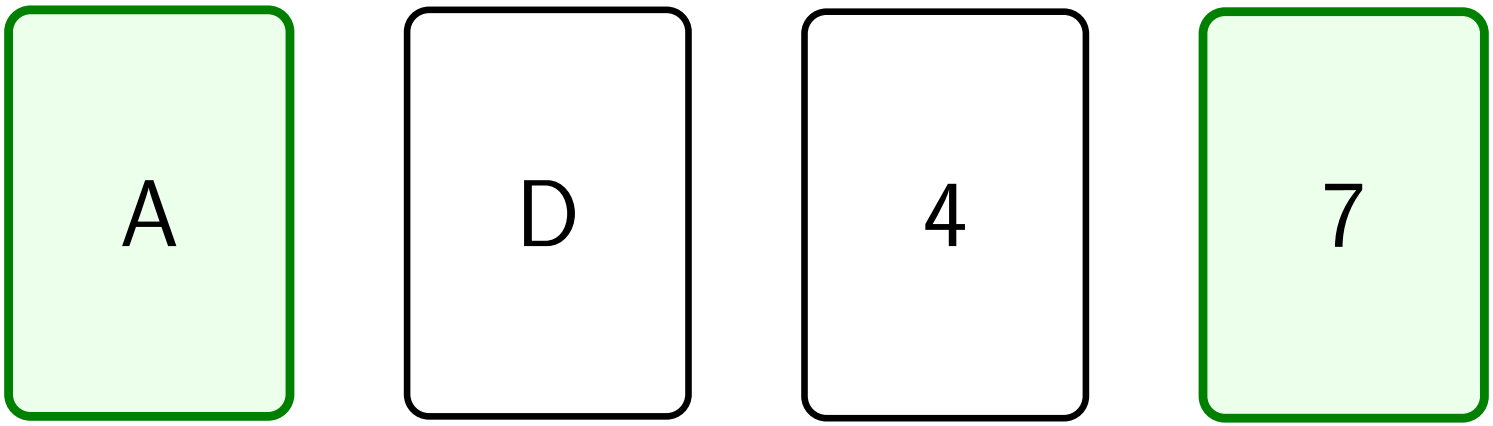
Visible	**Logical status**	**Could reveal $p \wedge \neg q$?**	**Turn?**
A	p	If back is odd ($\neg q$)	Yes
D	$\neg p$	No: even/odd doesn't matter	No
4	q	No: $p \wedge q$ and $\neg p \wedge q$ both OK	No
7	$\neg q$	If back is vowel (p)	Yes



Correct minimal test: turn A and 7.

Common Mistakes

- **Matching bias:** People tend to pick cards that *mention the items in the rule* (vowel, even) — i.e., A and 4.
- **Affirming the consequent:** Checking 4 corresponds to seeking evidence for $q \to p$, which is *not* implied by $p \to q$.
- **Falsification vs. verification:** To test a conditional, look for *counterexamples* ($p \wedge \neg q$), not confirming instances.



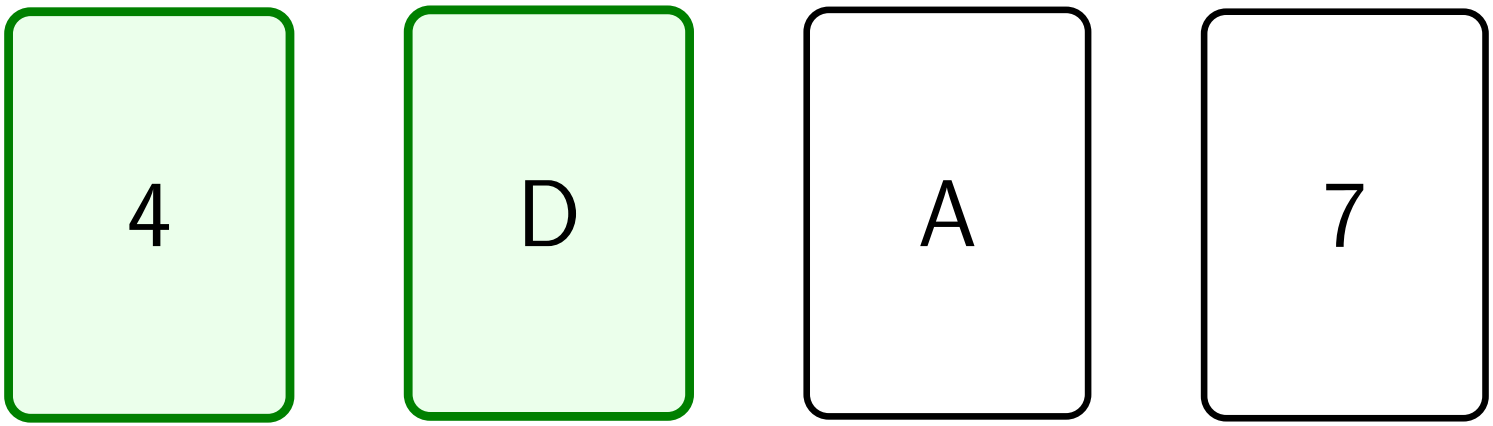
Reverse the conditional

Rule: "If a card has an even number, then it has a vowel." $(q \rightarrow p)$

Reverse the conditional

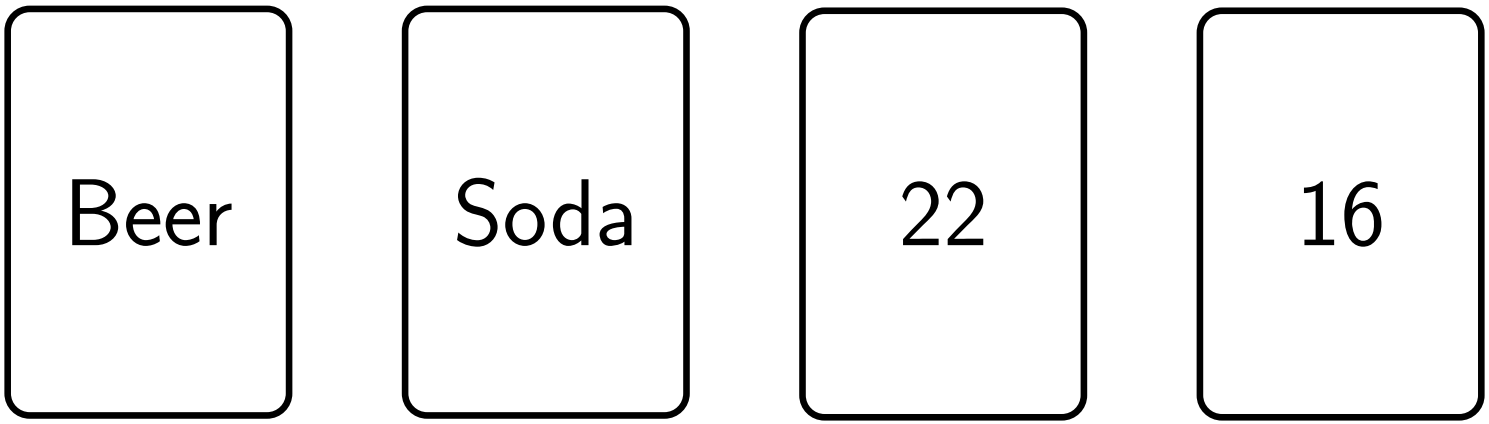
Rule: "If a card has an even number, then it has a vowel." $(q \to p)$

Falsify by finding $q \wedge \neg p \Rightarrow$ check $\boxed{4}$ and $\boxed{D}$.



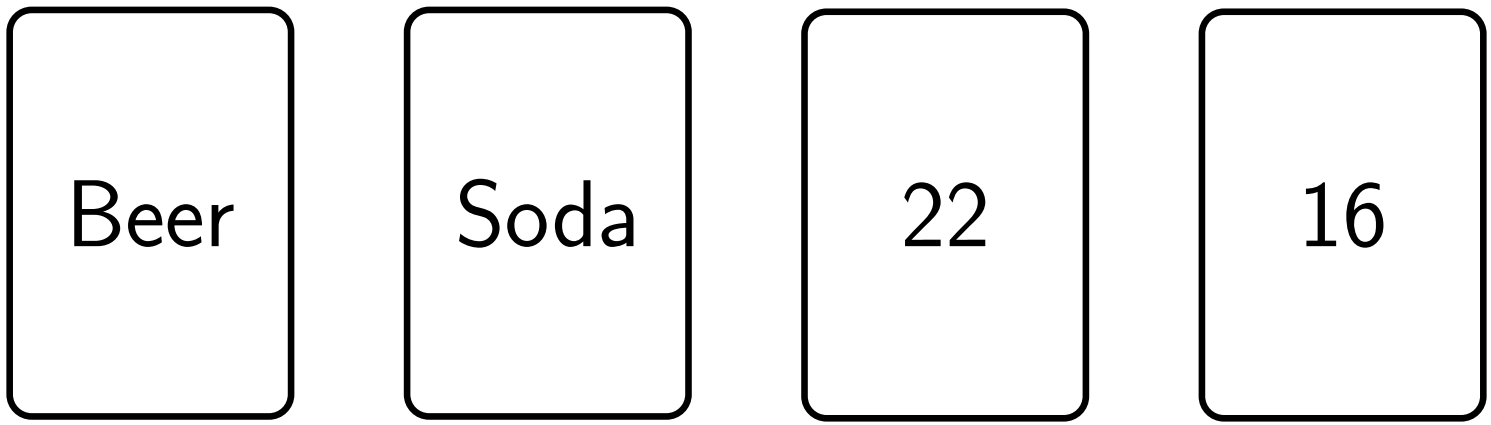
Interestingly, people get this right!

Rule: "If someone is **drinking beer**, then they **must be 21 or over**."



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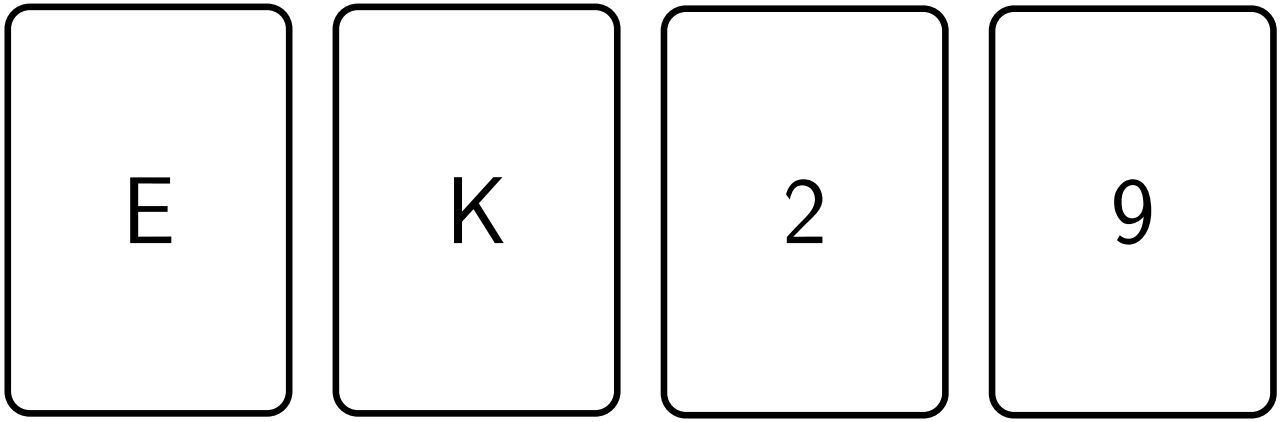


Correct: turn $\boxed{\text{Beer}}$ (p) and $\boxed{16}$ ($\neg q$).

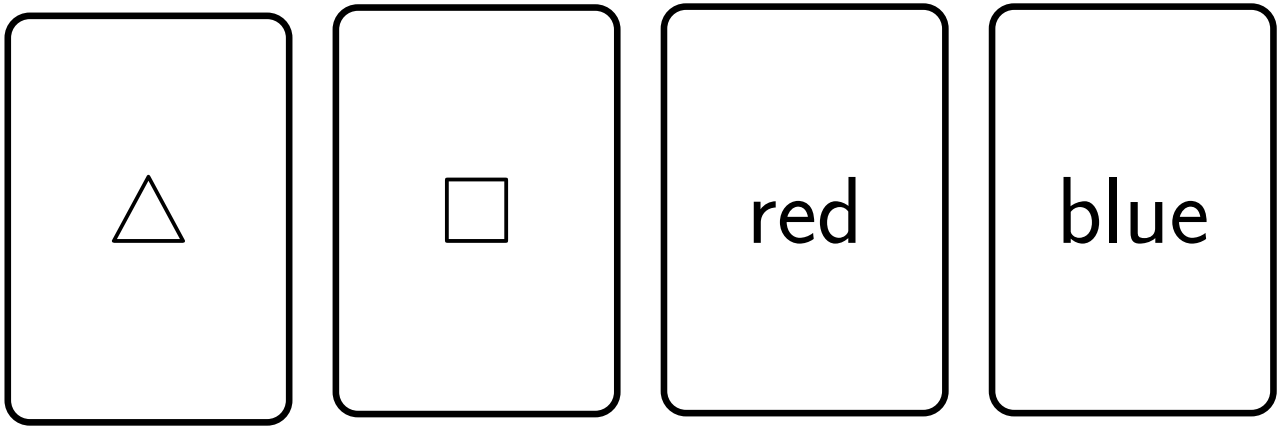
People typically do much better here than with A–D–4–7.

Class activity (2 min)

1. "If vowel then prime number."

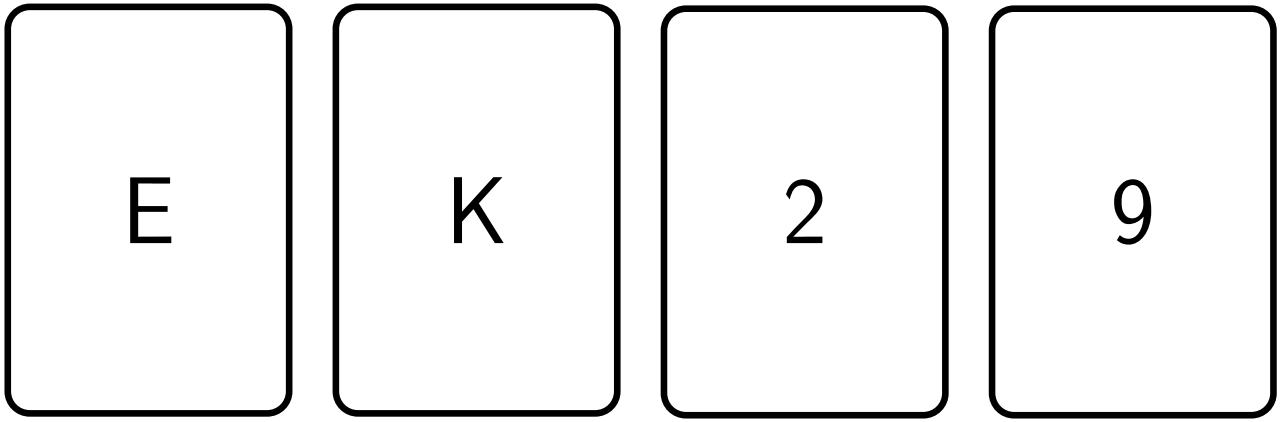


2. "If triangle then red."

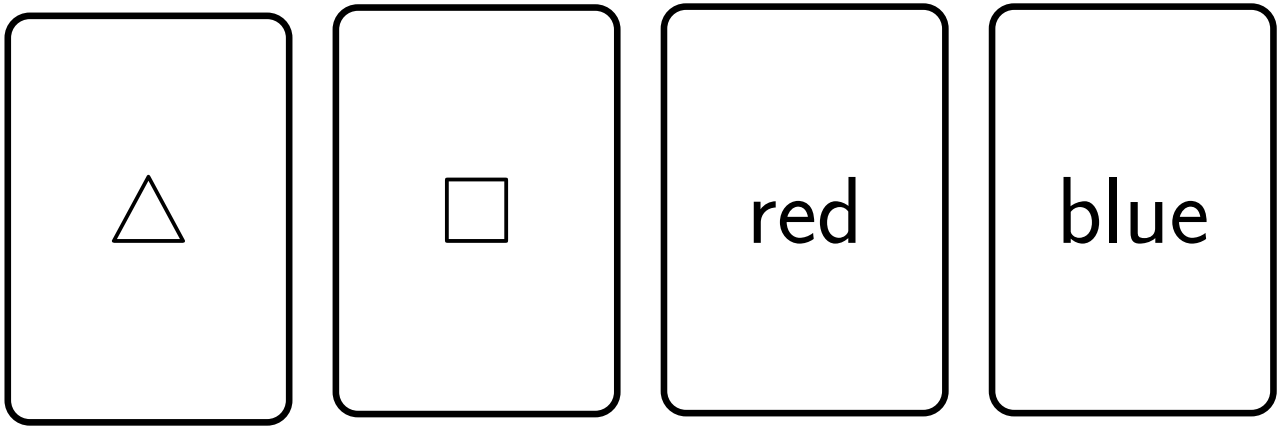


Class activity (2 min)

1. "If vowel then prime number."



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Reveal: p and $\neg q$ each time. (E and 9; triangle and blue.)

Logical Equivalence

Two propositions p and q are **logically equivalent** if they have the same truth value in all cases.

$$p \equiv q$$

Examples:

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Examples:

- $p \to q \equiv \neg p \vee q$
- $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$

De Morgan's Laws

Two Laws

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These show how negation distributes over "and" and "or."

Truth Table Proof of De Morgan's Law 1

Proof: $\neg(p \wedge q) \equiv (\neg p \vee \neg q)$

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	T	F	F
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

Columns 4 and 5 are identical so the two expressions are equivalent.

Truth Table Proof of De Morgan's Law 2

Proof: $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	T	F	F
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- $P(x) : x > 3$
- $P(5) \rightarrow$ True; $P(2) \rightarrow$ False.

Quantifiers

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Universal Quantifier

$\forall x\, P(x)$: "For all x, $P(x)$ is true."

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Universal Quantifier

$\forall x\, P(x)$: "For all x, $P(x)$ is true."

Existential Quantifier

$\exists x\, P(x)$: "There exists at least one x such that $P(x)$ is true."

Examples of Quantified Statements

- $\forall x \in \mathbb{R}$, $x^2 \geq 0$ – True.

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- $\forall x \in \mathbb{R}$, $x^2 \geq 0$ – True.
- $\exists x \in \mathbb{R}$, $x^2 = -1$ – False (in $\mathbb{R}$).
- $\forall n \in \mathbb{N}$, $n + 1 > n$ – True.

Negation of Quantified Statements

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Rules

$$\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x) \qquad \neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$$

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Example:

$$\neg(\forall x,\, x^2 \geq 0) \equiv \exists x,\, x^2 < 0$$

Combining Quantifiers

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- $\forall x \, \exists y \, (x + y = 0)$ – True ($y = -x$).
- $\exists y \, \forall x \, (x + y = 0)$ – False (no single y works for all x).

$$\forall x \, \exists y \, P(x, y) \not\equiv \exists y \, \forall x \, P(x, y)$$

Translating English to Logic

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- "All humans are mortal." $\rightarrow \forall x(H(x) \rightarrow M(x))$

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De Morgan's Laws for Quantifiers

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Example

$$\neg(\forall x,\ x > 0) \equiv \exists x,\ x \leq 0 \qquad \neg(\exists x,\ x < 0) \equiv \forall x,\ x \geq 0$$

Examples with Negations

Every student passed: $\forall x(S(x) \rightarrow P(x))$

Negation: $\exists x(S(x) \wedge \neg P(x))$

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Every student passed: $\forall x(S(x) \rightarrow P(x))$

Negation: $\exists x(S(x) \wedge \neg P(x))$

There is a prime even number: $\exists x(E(x) \wedge P(x))$

Negation: $\forall x(\neg E(x) \vee \neg P(x))$

Puzzle 1 — Tautology or Not

Decide whether the formula is a tautology, contradiction, or contingent:

$$\phi \;=\; (p \to q) \;\vee\; (q \to p).$$

Solution 1

$$(p \to q) \vee (q \to p) \equiv (\neg p \vee q) \vee (\neg q \vee p).$$

Associativity/commutativity:

$$= \neg p \vee p \vee \neg q \vee q = (\neg p \vee p) \vee (\neg q \vee q) = \top.$$

Hence ϕ is a **tautology**.

Puzzle 2 — Minimal Counterexample

Find a valuation (p, q, r) that satisfies

$$(\neg p \vee q) \ \wedge \ (\neg q \vee r) \ \wedge \ \neg(\neg p \vee r),$$

or prove none exists.

Puzzle 2 — Minimal Counterexample

Find a valuation (p, q, r) that satisfies

$$(\neg p \vee q) \ \wedge \ (\neg q \vee r) \ \wedge \ \neg(\neg p \vee r),$$

or prove none exists.

Solution 2 (Unsatisfiable):
Third conjunct $\neg(\neg p \vee r) \equiv p \wedge \neg r$. With p true, the first conjunct $(\neg p \vee q)$ forces q true. With q true, the second conjunct $(\neg q \vee r)$ forces r true. Contradiction with $\neg r$. So the set is **unsatisfiable**. $\square$

Puzzle 3 — Implication Intuition

Give a real-life sentence where $p \rightarrow q$ is true because p is false even though q is false.

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Give a real-life sentence where $p \to q$ is true because p is false even though q is false.

Solution 3:
Example: "If it rains on Mars today (p), the campus café is open (q)." If it does not rain on Mars (false p) and the café is closed (false q), then $p \to q$ is **true** (vacuously). □

Puzzle 4 — Quantifier Switcheroo ($\mathbb{Z}$)

Over $\mathbb{Z}$, decide truth:

$$\text{(A) } \forall x\, \exists y\ (x^2 + y^2 = 0) \qquad \text{(B) } \exists y\, \forall x\ (x^2 + y^2 = 0).$$

Puzzle 4 — Quantifier Switcheroo ($\mathbb{Z}$)

Over $\mathbb{Z}$, decide truth:

$$\text{(A) } \forall x\, \exists y\ (x^2 + y^2 = 0) \qquad \text{(B) } \exists y\, \forall x\ (x^2 + y^2 = 0).$$

Solution 4:

(A) False (unless complex numbers allowed, for each x, choose $y = ix$).
(B) False: no single y solves $x + y = 0$ for all x. $\square$

Puzzle 5 — Negate ε-δ

Negate and simplify:

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \; \big(|x - 2| < \delta \;\Rightarrow\; |x^2 - 4| < \varepsilon \big).$$

Puzzle 5 — Negate ε-δ

Negate and simplify:

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \; \big(|x - 2| < \delta \;\Rightarrow\; |x^2 - 4| < \varepsilon \big).$$

Solution 5:
Flip quantifiers and predicate:

$$\exists \varepsilon > 0 \; \forall \delta > 0 \; \exists x \; \big(|x - 2| < \delta \;\wedge\; |x^2 - 4| \geq \varepsilon \big).$$

□

Puzzle 6 — Translate to Logic

Formalize with $S(x)$: "x is a student", $Sub(x, a)$: "x submitted assignment a".

1. "Every student submitted at least one assignment."
2. "Some student submitted all assignments."
3. "No student submitted both A1 and A2."

Puzzle 6 — Translate to Logic

Formalize with $S(x)$: "x is a student", $Sub(x, a)$: "x submitted assignment a".

1. "Every student submitted at least one assignment."
2. "Some student submitted all assignments."
3. "No student submitted both A1 and A2."

Solution 6:

Puzzle 6 — Translate to Logic

Formalize with $S(x)$: "x is a student", $Sub(x, a)$: "x submitted assignment a".

1. "Every student submitted at least one assignment."
2. "Some student submitted all assignments."
3. "No student submitted both A1 and A2."

Solution 6:

1. $\forall x\,(S(x) \rightarrow \exists a\, Sub(x, a))$.

Puzzle 6 — Translate to Logic

Formalize with $S(x)$: "x is a student", $Sub(x, a)$: "x submitted assignment a".

1. "Every student submitted at least one assignment."
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Solution 6:

1. $\forall x\,(S(x) \rightarrow \exists a\, Sub(x, a))$.
2. $\exists x\,(S(x) \wedge \forall a\, Sub(x, a))$.

Puzzle 6 — Translate to Logic

Formalize with $S(x)$: "x is a student", $Sub(x, a)$: "x submitted assignment a".

1. "Every student submitted at least one assignment."
2. "Some student submitted all assignments."
3. "No student submitted both A1 and A2."

Solution 6:

1. $\forall x\,(S(x) \to \exists a\, Sub(x, a))$.
2. $\exists x\,(S(x) \wedge \forall a\, Sub(x, a))$.
3. $\forall x\,(S(x) \to \neg(Sub(x, A1) \wedge Sub(x, A2)))$.

□

Puzzle 7 — Additive Function on $\mathbb{N}$

Let $f : \mathbb{N} \to \mathbb{N}$ satisfy $f(m+n) = f(m) + f(n)$ for all m, n and $f(1) = 2$. Determine $f(n)$.

Puzzle 7 — Additive Function on $\mathbb{N}$

Let $f : \mathbb{N} \to \mathbb{N}$ satisfy $f(m+n) = f(m) + f(n)$ for all m, n and $f(1) = 2$. Determine $f(n)$.

Solution 7 (let's not forget Induction):

Claim: $f(n) = 2n$. Proof by induction: Base $n = 1$: $f(1) = 2 = 2 \cdot 1$.

Step: $f(n+1) = f(n) + f(1) = 2n + 2 = 2(n+1)$. $\square$

Puzzle 8 — Decide Truth over $\mathbb{R}$

Decide the truth of:

1. $\forall x \exists y \ (y^2 = x)$
2. $\exists y \forall x \ (y^2 = x)$
3. $\exists x \forall y \ (y^2 \geq x)$

Puzzle 8 — Decide Truth over $\mathbb{R}$

Decide the truth of:

1. $\forall x \exists y \ (y^2 = x)$
2. $\exists y \forall x \ (y^2 = x)$
3. $\exists x \forall y \ (y^2 \geq x)$

Solution 8:

(1) False: e.g., $x = -1$ has no real square root. (2) False: no single y squares to all x. (3) True: take $x = 0$ (indeed $y^2 \geq 0$ for all y). $\square$

Common Mistakes

- Reversing quantifier order.
- Negating without changing the quantifier.
- Mixing domains ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$, etc.).
- Assuming converse or inverse equivalence.